

σ -Modulated UCB Exploration for Bilateral Matching

1 σ -Modulated UCB Exploration

1.1 Limitation of Count-Based UCB1

The standard UCB1 selection rule for candidate-level exploration in the matching system is

$$\widetilde{M}_v = M_v + c \cdot \sqrt{\frac{\ln t}{n_v + 1}}, \quad (1)$$

where M_v is the analytical MapScore of candidate v , n_v is the number of times v has been selected up to round t , and $c > 0$ is the exploration constant. While Equation (1) guarantees sublinear regret $\mathcal{R}(T) = \mathcal{O}(\sqrt{T \log T})$, it imposes a uniform warmup: every unseen candidate receives an unbounded bonus when $n_v = 0$, forcing the system to try *every* candidate at least once before exploiting. With pool size $|\mathcal{V}| = 20$ and round budget $T = 60$, this warmup consumes 33% of the budget regardless of how informative the initial priors are.

This penalty contradicts a design goal of our system: *newcomers* (candidates with no platform history) deserve an upfront exploration bonus, but *established candidates with a brief absence from the current task* do not. Count-based UCB1 cannot distinguish these two cases because n_v measures only per-task selection history.

1.2 Proposed Rule

We replace Equation (1) with a σ -modulated variant that uses the candidate’s posterior uncertainty $\bar{\sigma}_v$ as an exploration intensity multiplier:

$$\widetilde{M}_v = M_v + c \cdot \bar{\sigma}_v \cdot \sqrt{\frac{\ln t}{n_v + 1}}, \quad (2)$$

where $\bar{\sigma}_v = \frac{1}{K} \sum_{k=1}^K \sigma_v^k$ is the average posterior standard deviation across v ’s K capability dimensions. Because σ_v^k shrinks monotonically as the platform accumulates feedback on v , $\bar{\sigma}_v$ encodes *platform-wide* information about v , not session-specific selection history.

Table 1 illustrates the resulting exploration intensity for representative candidate profiles at $c = 1$ and $t = 5$.

Three properties follow directly:

1. **Upfront semantics.** New users enter the pool with high $\bar{\sigma}_v$ from the L1 representation layer; they receive exploration bonus immediately in round 1, without requiring repeated non-selection.
2. **No forced warmup.** An established candidate with $n_v = 0$ in the current task does not receive an unbounded bonus, because $\bar{\sigma}_v$ is small. Veterans compete on M alone.

Candidate type	$\bar{\sigma}_v$	Bonus	Behavior
Brand-new (meta-knowledge source)	0.60	0.76	strong upfront bonus
New (implicit-knowledge source)	0.40	0.51	meaningful exploration
Established (explicit, few collabs)	0.15	0.19	mild bonus
Veteran (many past collaborations)	0.05	0.06	near pure greedy

Table 1: Exploration bonus under σ -modulated UCB at $n_v = 0$, $c = 1$, $t = 5$. Newcomers receive an order of magnitude more bonus than veterans.

3. **Self-decaying exploration.** Once a newcomer is selected and feedback is observed, the Bayesian posterior update shrinks σ_v^k , which in turn reduces $\bar{\sigma}_v$ and the future exploration bonus. The system does not over-explore.

1.3 Optional Sublinear Variant

For settings where newcomer prominence should be amplified beyond linear proportion, we propose a square-root variant:

$$\widetilde{M}_v = M_v + c \cdot \sqrt{\bar{\sigma}_v} \cdot \sqrt{\frac{\ln t}{n_v + 1}}. \quad (3)$$

This widens the gap between newcomer and veteran bonuses (Table 2).

$\bar{\sigma}_v$	Linear bonus (Eq. 2)	Square-root bonus (Eq. 3)
0.60	0.76	0.98
0.40	0.51	0.80
0.15	0.19	0.49
0.05	0.06	0.28

Table 2: Linear vs. square-root σ -modulation, $c = 1$, $t = 5$, $n_v = 0$.

1.4 Implementation

The change requires modifying a single line of the existing candidate selector:

```

1 # UCB1 (original)
2 bonus = self.candidate_ucb_c * math.sqrt(math.log(t) / (n_v + 1))
3
4 # sigma-modulated UCB
5 avg_sigma_v = mean(cap.sigma for cap in candidate.capabilities)
6 bonus = self.candidate_ucb_c * avg_sigma_v \
7         * math.sqrt(math.log(t) / (n_v + 1))

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2 Tiered Candidate Pool Design

2.1 Motivation: Coupling σ and μ Noise

A test pool with uniform $\sigma_v = 0.2$ and independently sampled μ noise fails to exercise the σ -modulated UCB mechanism. To distinguish σ -modulated UCB from count-based UCB1, the

dataset must contain candidates whose *both* μ -uncertainty and σ -uncertainty vary together—reflecting the real-world fact that a platform with little history on a user has both a noisy point estimate μ and low confidence σ .

We derive σ from a combination of the L1 knowledge source and prior collaboration history. Recall Proposition C of the proposal, which states that the posterior precision after n updates satisfies

$$\sigma_v(n) \leq \frac{\sigma_{\text{obs}}^{\text{source}}}{\sqrt{n+1}}, \quad (4)$$

where $\sigma_{\text{obs}}^{\text{source}} \in \{0.2, 0.4, 0.6\}$ for the explicit, implicit, and meta-knowledge sources respectively.

2.2 Tier Specification

We define four candidate tiers, each parameterized by an initial σ and a μ -noise standard deviation (Table 3).

Tier	Interpretation	σ_v	μ -noise	Pool fraction
veteran	Many past collaborations, well-known	~ 0.05	$\mathcal{N}(0, 0.02)$	25%
mid	2–3 past collaborations	~ 0.15	$\mathcal{N}(0, 0.10)$	25%
new_explicit	Recently joined, detailed self-report	~ 0.30	$\mathcal{N}(0, 0.20)$	25%
new_implicit	Recently joined, inferred from chat	~ 0.50	$\mathcal{N}(0, 0.30)$	25%

Table 3: Four-tier candidate pool design with coupled σ and μ noise.

For each candidate v assigned to tier τ , we draw

$$\sigma_v \sim \mathcal{U}(\sigma_\tau - \delta_\tau, \sigma_\tau + \delta_\tau), \quad (5)$$

$$\mu_v^{\text{init},k} = \text{clip}(\mu_v^{\text{true},k} + \varepsilon_k, 0, 1), \quad \varepsilon_k \sim \mathcal{N}(0, \eta_\tau^2), \quad (6)$$

where $\sigma_\tau, \eta_\tau, \delta_\tau$ are the tier parameters from Table 3 and a small jitter δ_τ (e.g., 0.02–0.08) prevents identical σ values within a tier.

2.3 Expected Algorithmic Behavior per Tier

Under σ -modulated UCB with $c = 1$, $t = 5$:

- **Veterans** ($\sigma = 0.05$): exploration bonus ≈ 0.06 ; the system ranks them on near-true μ since their μ -noise is also small.
- **Mid-experience** ($\sigma = 0.15$): bonus ≈ 0.19 ; mild exploration if base M is competitive.
- **New explicit** ($\sigma = 0.30$): bonus ≈ 0.38 ; $6\times$ a veteran’s bonus, ensuring meaningful exploration despite noisy μ .
- **New implicit** ($\sigma = 0.50$): bonus ≈ 0.63 ; $10\times$ a veteran’s bonus. This is the tier most likely to harbor underestimated high-ability candidates (“hidden gems”) and the strongest test of the anti-Matthew-effect property.